

Brazil’s soy land footprint is shifting toward China and toward the agricultural frontier

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Abstract

TODO — Methods-only working document. The full abstract lives in the main manuscript. Placeholder so the file compiles standalone.

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1 Methods

1.1 Overview and notation

We couple a subnational (municipality-level) reconstruction of Brazil’s soy supply chain with the global FABIO physical multi-regional input–output (MRIO) framework [1], and attribute the resulting land-use footprint back to municipalities of origin and forward to countries of final consumption, annually for 2001–2022 (on the FABIO v2 backend from 2010 and the v1.1 backend before; Section 1.7.1). The construction proceeds in six stages: (i) municipal soy accounting, assembling production, trade and domestic-use quantities for every Brazilian municipality (Section 1.2); (ii) livestock-system disaggregation and feed demand (Section 1.3); (iii) reconciliation of the municipal totals with national commodity balances (Section 1.4); (iv) allocation of physical flows between municipalities by cost-minimizing optimal transport (Section 1.5); (v) linkage of origin municipalities to importing countries, with re-export disentangling (Section 1.6); and (vi) nesting into FABIO and land-use footprint accounting (Sections 1.7–1.8). The resulting flows are validated against the independent Trase supply-chain benchmark and a pure downscaling baseline (Section 1.9). The subnational tracing framework generalizes the single-year (2013) open model of

Trsek [2] to an annual series; a script-level mapping of each subsection to the released pipeline code is provided with the repository. Figure 1 summarizes the full pipeline.

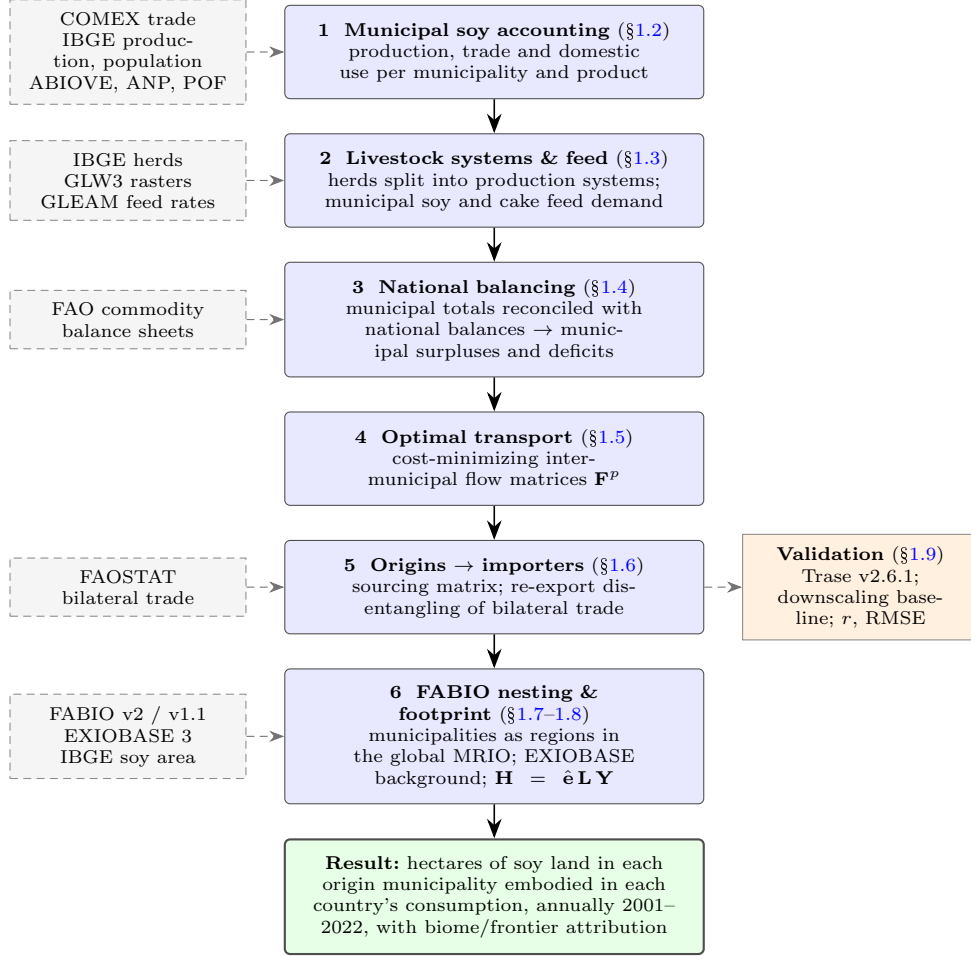


Fig. 1 Workflow of the study. Solid boxes are the six model stages (Sections 1.2–1.8); dashed boxes are the main data inputs at each stage; the orange box is the independent validation of the reconstructed flows (Section 1.9). Stages 1–5 reconstruct Brazil’s subnational soy supply chain; stage 6 embeds it in the global FABIO/EXIOBASE input–output framework and computes the land-use footprint.

Notation.

Throughout, m and n index Brazil’s $\sim 5,570$ municipalities, $p \in \{\text{bean, oil, cake}\}$ the three soy products, and i, j countries. National annual totals from the FAO commodity balance sheet are written $\text{CBS}[p, \text{use}]$, e.g. $\text{CBS}[\text{oil, food}]$ for the national food use of

soy oil. Matrices and vectors are set in bold, $\text{diag}(\cdot)$ denotes a diagonal matrix and $\mathbf{I}$ the identity; a hat ($\hat{\mathbf{e}}$) diagonalizes a vector.

The mass-conserving allocation operator.

A single operation recurs throughout the subnational reconstruction: a national total T is distributed across municipalities in proportion to an observable weight w_m ,

$$q_m = \frac{w_m}{\sum_n w_n} T, \quad \text{so that} \quad \sum_m q_m = T. \quad (1)$$

The stages below differ mainly in the choice of weight w_m ; the released pipeline checks each allocation against its national total at run time and halts on any violation. Table 1 collects the weights used for the domestic-use items.

1.2 Municipal soy accounting

1.2.1 Municipal supply, trade and infrastructure

Municipality-level exports and imports of the three soy products are read from COMEX customs records, aggregated from monthly to annual tonnages and matched to IBGE municipality identifiers. Exports declared without a municipality of origin are redistributed across the declared exporters of the same product and destination in proportion to their declared volumes, thereby conserving national totals. Undeclared imports are negligible and are not redistributed. Installed crushing and refining capacity (ABIOVE) is divided equally among each state’s active plants and summed per municipality. Grain storage capacity is aggregated from georeferenced warehouse records, and biodiesel capacity (ANP) is attributed to soy according to the soy feed-stock share of each macro-region. All geometries are projected onto a national planar reference system (SIRGAS 2000 / Brazil Polyconic, EPSG:5880) and each municipality is represented by its administrative centre. Straight-line distances between administrative centres are used to calculate transport costs in Section 1.5.

1.2.2 Domestic-use estimation

Every item intended for domestic use in the national commodity balance is distributed across municipalities using the allocation operator (see equation (1)) and the weights in Table 1. The national soybean crush is allocated in proportion to crushing capacity. Municipal oil and cake output then follow from the crush, with conversion yields derived each year from the national balance itself (the ratio of national oil and cake production to beans crushed is approximately 20% oil and 75% cake, with a 5% processing loss, consistent with values in the literature). The food use of beans and oil is allocated by household soy oil consumption, which is calculated as the state-level per capita acquisition rate (POF) multiplied by the municipal population. Other uses, such as biodiesel production, are allocated according to soy-attributable biodiesel capacity. Seed use is allocated according to soybean production, and stock changes are allocated according to storage capacity for beans and to municipal oil and cake production for the processed products. A net national stock addition is carried on the use side,

whereas a net national withdrawal is treated as a supply component (see Eq. 2) and distributed across municipalities by those same stock weights, but added to municipal supply rather than subtracted from use. This mirrors the commodity-balance identity in which a drawdown is a genuine source of current-year supply and ensures that every municipal use remains non-negative, as required by the re-export inversion in Section 1.6.2. A storage-based split of a drawdown on the use side would instead result in the net use of high-storage, low-use municipalities being negative.

Table 1 Domestic-use items as instances of the allocation operator (1): each item is a national commodity-balance total distributed across municipalities by the stated weight w_m .

Use item	Allocation weight w_m
Processing (crush)	installed crushing capacity
Food (beans, oil)	per-capita soy-oil acquisition $\times$ population
Other (biodiesel)	soy-attributable biodiesel capacity
Seed	soybean production
Stock change	storage capacity (beans), production (oil, cake); addition $\rightarrow$ use, withdrawal $\rightarrow$ supply

1.3 Livestock systems and feed demand

As soy feed intake varies significantly across production systems, IBGE municipal herd counts are divided into the following categories: extensive and intensive chickens; extensive, intensive and industrial pigs; and grassland- and mixed-based cattle and buffalo, with each category divided into dairy and meat. System shares are obtained using area-weighted zonal statistics of the 2010 FAO gridded livestock rasters (GLW3) and the GLEAM ruminant production system map for each municipality. Where a municipality reports a herd but had no gridded animals in 2010, the state mean share is used. Dairy herds are identified by milked cow counts and feedlot cattle, which are only reported in the agricultural census, are extrapolated from 2006 using the national confinement growth rate while holding the municipal pattern fixed. The full split formulas are given in Appendix A. Each species’ subsystems are verified to partition its reported herd exactly.

Feed demand is calculated by multiplying each herd by its GLEAM per-animal intake, i.e. annual dry-matter intake multiplied by the proportion sourced from soy-beans and soy cake, converted to as-fed tonnes at 88% dry matter content. One national factor per product then rescales the resulting bottom-up totals to the FAO national feed figures (an application of equation (1) with the bottom-up municipality $\times$ system feed matrix as weight), preserving the spatial and cross-system distribution while matching the national constraint. Municipal feed demand for beans and cake is the sum over systems. Holding the 2010 system shares constant over the study period is a simplification: herd sizes remain annual, misclassification affects only how a municipality’s herd is split across intake intensities, and the subsequent rescaling of all bottom-up

feed to the national FAO totals bounds the aggregate error. The residual uncertainty grows with distance from the 2010 reference year. Figure 2 sketches the construction.

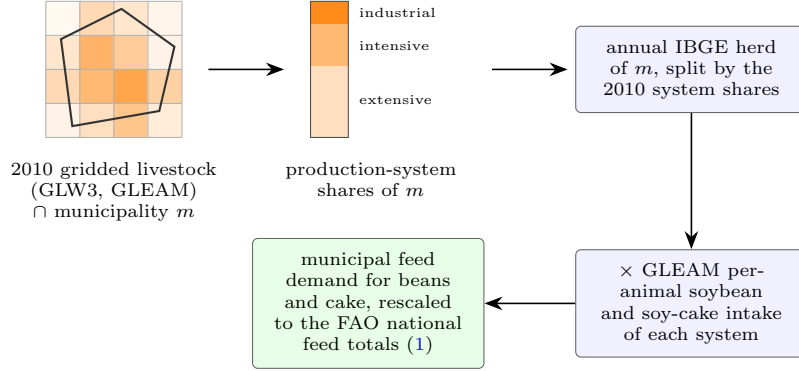


Fig. 2 From herds to municipal soy feed demand. Municipal herd counts (annual) are split into production systems using the shares implied by the 2010 gridded-livestock rasters within each municipality; each system’s herd is multiplied by its per-animal soy and soy-cake intake, and the bottom-up totals are rescaled to match the national feed use reported by FAO.

1.4 National balancing and trade reconciliation

For each product, the national commodity balance imposes the supply–use identity

$$\underbrace{\text{production} + \text{imports} + \text{stock withdrawal}}_{\text{supply}} = \underbrace{\text{exports} + \text{food} + \text{feed} + \text{seed} + \text{processing} + \text{other}}_{\text{use}}. \quad (2)$$

First, municipal exports and imports are harmonised with the FABIO/FAOSTAT bilateral trade classification. Destinations absent from the bilateral trade list are collapsed into a rest-of-world aggregate. This step aligns classifications and applies no rescaling. Any remaining supply–use residual at the national level is absorbed into the stock term and each balance item is then proportionally rescaled by equation (1) using the item’s municipal distribution as the weight. This ensures that every municipal aggregate matches its national total exactly. Consistent with the supply–use identity (Equation (2)), a net national stock withdrawal is carried out on the supply side: the drawdown is distributed across municipalities by the stock weights (storage capacity for beans and production for oil and cake) and added to municipal supply. This ensures that every municipal use remains non-negative (additional years are unchanged and carried on the use side). As the model is a single annual snapshot, tonnes supplied from a drawdown, which were physically grown in earlier years, inherit the current year’s soy-area intensity (Section 1.8). This approximation only affects the small share of throughput met from net stock changes. Finally, each municipality’s product balance

defines its surplus or deficit:

$$\text{surplus}_m^p = \max(\text{supply}_m^p - \text{use}_m^p, 0), \quad \text{deficit}_m^p = \max(\text{use}_m^p - \text{supply}_m^p, 0), \quad (3)$$

These then determine the inter-municipal flows of Section 1.5. The positive-part decomposition is lossless: surplus minus deficit equals the signed net position of every municipality, and the two totals match by construction after the above rescaling, as required by the balanced transportation problem. At the national level, the corresponding residual cannot be resolved into flows. There, it is retained as Fabio’s signed balancing column, which re-enters the model as a dedicated final-demand category (see Sections 1.7.2 and 1.8.1).

1.5 Subnational transport allocation

Municipal surpluses and deficits are connected by *cost-minimizing optimal transport* rather than a gravity or distance-decay model, on the premise that bulk-commodity logistics minimize freight cost and that freight cost increases with distance. The cost metric is the straight-line distance between municipal seats, which requires no network parameters that would be unobservable for parts of the study period; its adequacy is tested against the independent benchmark (Section 1.9). The multimodal network data (roads, rail, waterways, ports; Data) support an experimental cost-based variant that is carried in the repository but not used for the reported results. **[ADD MULTIMODAL VARIANT: describe the multimodal cost-based transport model here once it is runnable, and report the sensitivity of the footprints to the cost metric.]** For each product p , the inter-municipal flow matrix $\mathbf{F}^p$ solves the balanced transportation problem

$$\mathbf{F}^p = \arg \min_{\mathbf{F} \geq 0} \sum_{m,n} F_{mn} d_{mn} \quad \text{s.t.} \quad \sum_n F_{mn} = \text{surplus}_m^p, \quad \sum_m F_{mn} = \text{deficit}_n^p, \quad (4)$$

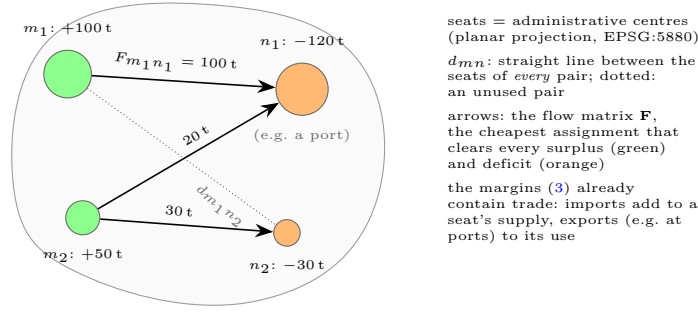
where d_{mn} is the straight-line distance between the seats of municipalities m and n (the two margins are balanced beforehand by proportionally scaling the larger one). The problem is solved exactly by the network-simplex algorithm (R package `transport` [3]), separately for beans, oil and cake. The solution is a deterministic, mass-conserving flow matrix whose margins reproduce the municipal surpluses and deficits by construction. Figure 3 contrasts this allocation with the pure-downscaling baseline used for validation (Section 1.9.1) on a stylized example.

1.6 From origins to consumers

1.6.1 Linking origin municipalities to importing countries

Customs data attach two labels to every exported tonne: the municipality that shipped it and the country that first imported it. Neither is the label the footprint needs, which is the municipality that grew the soy and the country that finally consumes it. This section corrects both, first backwards within Brazil (this part), then forwards across borders (Section 1.6.2).

a subnational model: cost-minimizing optimal transport (4)



b downscaling baseline: production shares only (1)

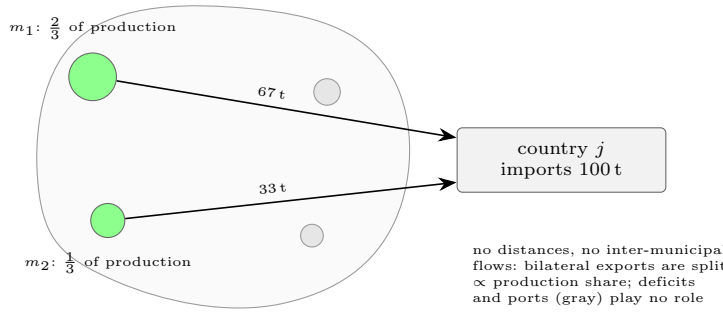


Fig. 3 Two ways of attributing trade to origin municipalities, on the same stylized geography. **a**, The subnational model: each municipality is represented by its administrative centre (seat), and d_{mn} is the straight-line (Euclidean) distance between the seats of every pair of municipalities in the planar national projection. The flow matrix $\mathbf{F}$ (arrows) is the solution of the balanced transportation problem (4) — the cheapest mass-conserving assignment of surpluses (green, circle area proportional to tonnage) to deficits (orange): here m_2 serves the nearby small deficit n_2 and the remainder flows to n_1 ; the pair $m_1 n_2$ (dotted) is priced but not used. International trade is embedded in the margins (Eqs. 2 and 3): a municipality's imports add to its supply and its exports add to its use, so an exporting port municipality appears simply as a large deficit (n_1) that attracts inter-municipal flows — no separate trade arrows are needed. **b**, The pure-downscaling baseline used for validation (Section 1.9.1) ignores the domestic network entirely: exports to each destination are split across municipalities in proportion to their production share (here $\frac{2}{3} \times 100 = 67$ t and 33 t), i.e. the allocation operator (1) applied directly to bilateral exports; deficits, ports and inter-municipal flows (gray) play no role.

The transport flows of Section 1.5 are the tracing device, after one completion: they record only what moves between municipalities, so the part of each municipality's supply that stays and is used at home is added on the diagonal,

$$\tilde{F}_{mn}^p = F_{mn}^p + \mathbf{1}[m = n] (\text{supply}_m^p - \text{surplus}_m^p), \quad (5)$$

so that every tonne a municipality handles is accounted for, whether it travelled or stayed: the rows of $\tilde{\mathbf{F}}^p$ sum to total municipal supply, its columns to total municipal use. Figure 4 illustrates the construction and the sourcing shares derived from it.

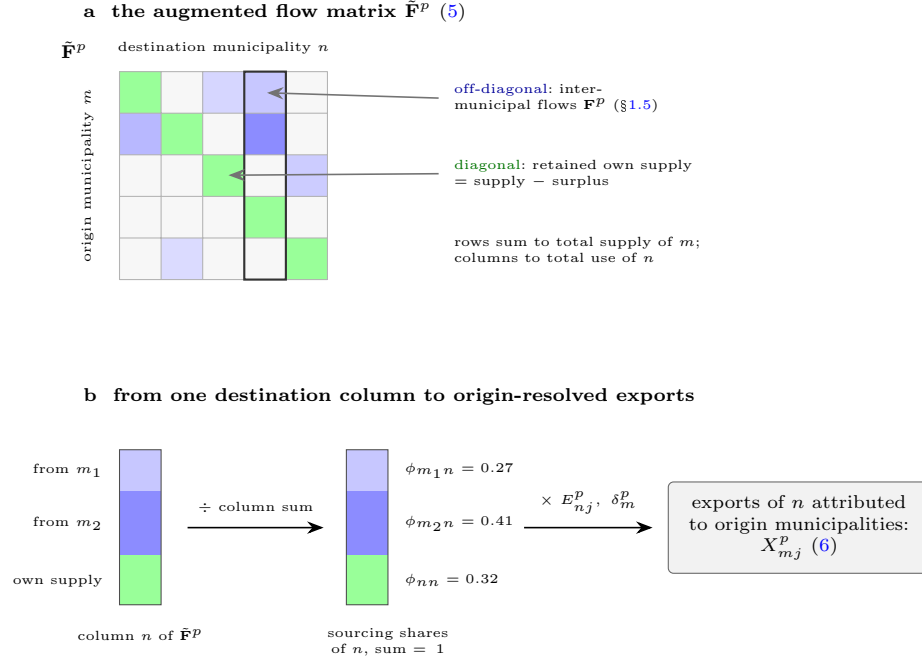


Fig. 4 Completing the flow matrix and reading off sourcing shares. **a**, The inter-municipal flows $\mathbf{F}^p$ of Section 1.5 occupy the off-diagonal (blue); the diagonal (green) adds each municipality’s retained own supply (5), so rows sum to total municipal supply and columns to total municipal use. **b**, One destination column (framed in **a**), divided by its sum, gives the sourcing shares ϕ_{mn}^p : the composition of municipality n ’s use by origin, including n ’s own retained supply. Weighting n ’s observed exports E_{nj}^p by these shares and by the domestic-origin share δ_m^p yields the origin-resolved exports X_{mj}^p of Eq. (6).

Every exported tonne is then unwound to its origin,

$$X_{mj}^p = \delta_m^p \sum_n \phi_{mn}^p E_{nj}^p, \quad (6)$$

where E_{nj}^p are the observed exports of municipality n to country j ; the sourcing share $\phi_{mn}^p = \tilde{F}_{mn}^p / \sum_{m'} \tilde{F}_{m'n}^p$ is the fraction of n ’s total use that originated in municipality m (including $m = n$); and the domestic-origin share $\delta_m^p = (\text{supply}_m^p - \text{imports}_m^p) / \text{supply}_m^p$ strips out quantities of foreign origin (withdrawal-supplied tonnes count as domestic under the supply-side stock treatment; Section 1.2.2). Local consumption is traced with the same formula through a synthetic domestic column added to the export table (for beans net of the crush, which is traced through the oil and cake chains instead), and oil and cake are traced one step further back, to the bean-growing municipalities,

by premultiplying their origin-to-destination matrices with the bean sourcing shares and the bean domestic-origin share.

1.6.2 Re-export disentangling

To attribute internationally traded soy to its true origin, bilateral trade is disentangled with a Leontief-type inversion. For each commodity, let $\mathbf{T}$ be the bilateral trade matrix, with Brazil replaced by its municipalities for the soy products, and $\mathbf{u}$ the vector of total use by region. Normalizing trade by destination use and summing the infinite re-export chain,

$$\mathbf{R} = \mathbf{T} \hat{\mathbf{u}}^{-1}, \quad \mathbf{L}_{\text{trade}} = (\mathbf{I} - \mathbf{R})^{-1} = \sum_{k=0}^{\infty} \mathbf{R}^k, \quad (7)$$

where the entry R_{ij} is the share of region j 's total use supplied by imports from region i , so that $\mathbf{R}^k$ carries the flows that have been re-exported k times and the inverse sums chains of every length. Retaining only the quantities produced domestically (via each region's domestic supply share), scaled by each destination's domestic use, yields the origin-to-consumption matrix. The entry in this matrix gives the quantity produced in region (or municipality) i and ultimately consumed in region j . The inversion assumes proportional mixing of domestic production and imports within each region's use, the standard assumption of physical trade-linkage accounting [4]. At the municipal level, this is stronger than at the country level in an MRIO, since it also governs which producing municipalities supply each destination. The export leg is nonetheless observed at the municipality–country level, so destination differences that operate through port choice are retained, and the assumption binds only to sourcing within a port's hinterland and to the international re-export chain. Origin-specific contracting (e.g. the Amazon Soy Moratorium and certified zero-deforestation chains) is not resolved; its bias is bounded in Section 1.9.4. Its row and column sums recover each region's domestic supply and use.¹ The inversion is computed exactly by a sparse solver. If the sparse factorisation fails numerically, a dense solve is attempted. Only as a last resort is a small diagonal ridge applied to lift a genuine unit eigenvalue of $\mathbf{R}$ (which arises when a commodity's recorded global trade is entirely re-exported). The years in which the ridge is required are documented with the released code. Figure 5 illustrates the correction for a stylized example.

1.7 Nesting into FABIO

The reconstructed municipal soy flows are embedded in the FABIO v2 physical MRIO, in which a *process* is a region $\times$ commodity unit. Brazil's national soy commodities and processes are removed and replaced by their municipal counterparts throughout, so that municipalities enter the global model as additional regions. Figure 6 illustrates the nesting and the role of the Leontief inverses used below.

¹For soy products, the regional dimension comprises non-Brazilian trading countries and Brazilian municipalities, with Brazil as a whole fully replaced by its municipalities. This replacement is enforced throughout the processing of the trade data so that no national Brazilian record re-enters the inversion.

a observed trade: country of *first import*



b after re-export disentangling (7): country of *consumption*

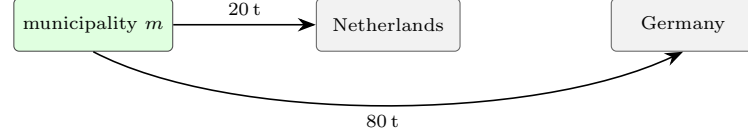


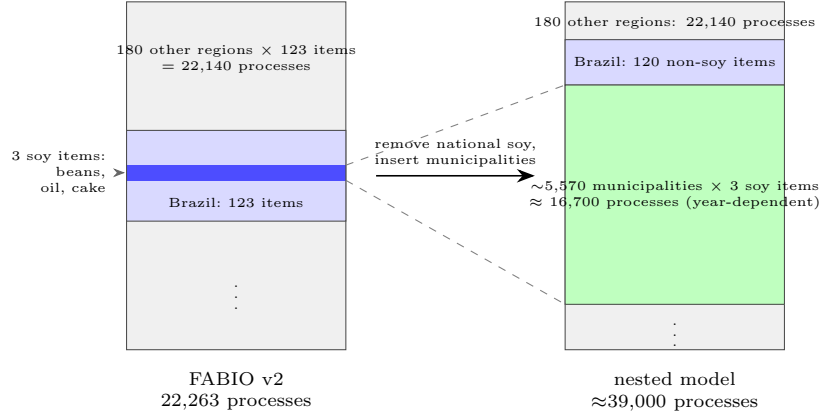
Fig. 5 Re-export disentangling. **a**, Customs records attribute all 100 t exported by municipality m to the Netherlands, the country of first import, although 80 t are re-exported and consumed in Germany. **b**, The trade inversion (7) sums re-export chains of any length and reattributes each tonne to its country of final consumption, leaving only the 20 t actually consumed in the Netherlands attributed there.

1.7.1 Temporal coverage: the v2 and v1.1 backends

The FABIO v2 build only exists for 2010 onwards. For the years 2001–2009, the pipeline therefore switches to using four inputs from the FABIO v1.1 build, which covers this period: the full bilateral trade detail (1986–2019), the land-use extension (1986–2013), the feedstock optimisation results (1961–2019) and the balanced bilateral trade and commodity balance sheets from the v1 release (1986–2013 and 1961–2019). The v1 balance sheets are used because the refreshed v2 balance file only contains placeholder rows for the pre-2010 years (for soybeans, production equals feed, and imports, exports, processing, and food are all zero up to and including 2009), which would divert all trade into stock changes. All version bridges map through stable FAO item codes across v1.1 and v2, but not through FABIO commodity codes, which differ between the versions. The v1.1 land-use extension is supplied as a long table and converted to the per-process intensity vector through the same item-code mapping. The extensions also differ semantically: v2 provides a cropland stressor, whereas the v1.1 extension reports total agricultural land use, including pasture for livestock processes. Soy processes are cropland under both definitions, so the soy land footprint remains unaffected. However, non-soy land intensities cannot be compared across vintages.

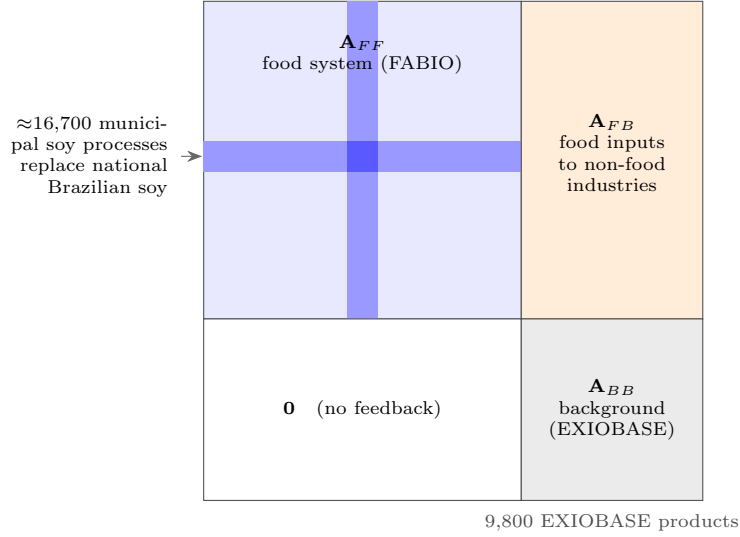
The seam between the two backends is quantified by computing the overlap years 2010–2013, which all inputs of both backends cover, through both chains end to end. Where the comparison is clean the seam is small and stable: totals differ by -6% in 2011 and -8% in 2013 (v1.1 low), the cross-country correlations of consumer footprints are 0.988 and 0.992, and China, the largest consumer, agrees within $\pm 3\%$ in both years. The two remaining overlap years illustrate why we disclose the seam rather than correct for it: the 2012 v1.1 build halts at the re-export conservation check (a

a



each municipal soy process carries its own production and soy area (§1.2),
feed use (§1.3) and origin-resolved trade (§§1.5–1.6); block heights not to scale

b $\approx 39,000$ processes (FABIO + municipal soy)



Leontief inverses: $\mathbf{L}_F = (\mathbf{I} - \mathbf{A}_{FF})^{-1}$ — supply chains inside the food system; $\mathbf{L}_B = (\mathbf{I} - \mathbf{A}_{BB})^{-1}$ — supply chains of the background economy; $\mathbf{L}_{FB} = \mathbf{L}_F \mathbf{A}_{FB} \mathbf{L}_B$ — links non-food consumption (e.g. biodiesel) back to food-system land use.

Fig. 6 Nesting municipal soy into the global input-output system. **a**, Insertion of municipal Brazil into FABIO: the three national Brazilian soy commodities are removed from the process list and replaced by one soy process set per municipality, populated with the municipal accounts of Sections 1.2–1.6; all other regions, and Brazil’s remaining commodities, are untouched. **b**, Structure of the hybrid technical-coefficient matrix. Brazilian municipalities enter the FABIO block $\mathbf{A}_{FF}$ as additional regions (dark band), replacing national Brazilian soy; the EXIOBASE background economy $\mathbf{A}_{BB}$ receives food-system inputs through the linking block $\mathbf{A}_{FB}$ but does not feed back, so the system is block-triangular.

strong stock-drawdown year on the v1 balance sheets), and 2010 diverges more widely (-31% overall, China -22% , cross-country correlation 0.964), a disagreement between the balance-sheet vintages in that year rather than a property of the footprint machinery. With only two clean overlap points, a splice adjustment of the pre-2010 levels would rest on weaker evidence than the $\pm 8\%$ seam it removes, so pre-2010 levels are reported unadjusted. The one systematic difference across all overlap years is Brazil’s domestic footprint, about a third lower on v1.1 because the v1 balance sheets book most domestic soy use in an unspecified final-demand category; cross-vintage statements about Brazil-domestic and about end-use composition are therefore restricted to 2010 onwards. Two further pre-2010 caveats apply. First, in 2004 the southern-drought supply shock is partly absorbed by the balancing residual with opposite regional signs (positive in the centre-west, negative in the south), so the *municipal* attribution within Brazil is distorted in that year, while the national total and the consumer-country distribution remain in line with the neighbouring years. Second, year 2000 is excluded from the reported series pending resolution of a demand-side allocation issue in that year’s build.

1.7.2 From municipal accounts to the nested MRIO

The nested model is assembled in three stages.

First, supply and use tables are constructed for every region, with each municipality treated as a region in its own right. The construction follows the published FABIO procedure [1]: joint production is split according to process production shares, and a parallel monetary table is derived from bilateral-trade unit prices (winsorized and gap-filled), with all municipalities inheriting Brazil’s national price level. The single substantive departure concerns feed: the municipal soybean and soy-cake feed demands of Section 1.3 replace FABIO’s national feed use and are assigned to the livestock processes of the respective municipality. All other use categories remain as in the national model.

Second, the regional tables are linked through trade. The intermediate use and final demand of every commodity in every destination are distributed across supplying regions in proportion to the re-export-free trade flows of Section 1.6.2; grazing and stock withdrawals are treated as fully domestic, since neither is traded internationally. Municipal livestock use and final demand are subsequently re-aggregated to national Brazilian columns, whereas municipal soy production remains disaggregated, so that municipal detail is preserved precisely where the footprint originates.

Third, the linked tables are condensed into a symmetric commodity-by-commodity MRIO under the industry-technology assumption [5], and the Leontief inverse $\mathbf{L}$ is computed, in parallel on a physical (mass) and a monetary (value) basis. Any remaining supply–use residual is routed to a dedicated balancing final-demand column, and spurious negative coefficients arising from this residual are set to zero. In addition, a productiveness safeguard rescales any column of $\mathbf{A}$ whose sum reaches one, a configuration in which intermediate use would equal or exceed gross output and $\mathbf{I} - \mathbf{A}$ would become singular, to just below one. Columns of this kind are inherited from the FABIO base and arise in several thousand national food-system processes of countries other than Brazil, under both backends; the Brazilian municipal soy processes

that carry the land footprint are never affected. The sensitivity of the headline footprints to the cap threshold, together with a per-year account of the capped columns, is reported in the uncertainty analysis (Section 1.9.4).

Non-food uses of soy (e.g. biodiesel) leave the food system and enter the wider economy, supplied by EXIOBASE 3 [6] (product-by-product, 49 regions $\times$ 200 products): concordances map FABIO commodities and countries to EXIOBASE products and regions, and each destination’s “other use” is split across EXIOBASE sectors in proportion to that region’s product technology, yielding a linking block $\mathbf{A}_{FB}$. With no physical feedback from the background economy into FABIO, the hybrid system is block-triangular and its cross-system Leontief block is simply

$$\mathbf{L}_{FB} = \mathbf{L}_F \mathbf{A}_{FB} \mathbf{L}_B, \quad (8)$$

with $\mathbf{L}_F$ the FABIO and $\mathbf{L}_B$ the EXIOBASE Leontief inverse; $\mathbf{L}_{FB}$ carries final demand for non-food products back to FABIO land use.

1.8 Land-use footprint accounting

1.8.1 Footprint calculation

The direct land use of each process is its harvested soy area (from IBGE, for municipal soy processes) or the FABIO cropland extension (for national processes); dividing by total output gives the land intensity $e_i = \text{land}_i/x_i$ of every process i . The footprint is the standard intensity–Leontief–demand product

$$\mathbf{H} = \hat{\mathbf{e}} \mathbf{L} \mathbf{Y}, \quad (9)$$

where $\mathbf{Y}$ is final demand aggregated to consumer countries and H_{ij} is the hectares of soy land in origin municipality i embodied in country j ’s consumption.² The food system ($\mathbf{L}_F$) and the non-food background (via $\mathbf{L}_{FB}$) contribute additively to each consumer’s total. Footprints by consumed product are obtained by restricting $\mathbf{Y}$ to selected commodity groups (e.g. dairy, meat) before applying (9).

1.8.2 Origin attribution and grid refinement

Each municipality is assigned to a biome by intersecting its interior point with the IBGE 1:250,000 biome layer (points falling outside every biome polygon are assigned the nearest biome). The agricultural *frontier* is defined as the Amazon biome plus MATOPIBA (operationalized as the Cerrado biome within Maranhão, Tocantins, Piauí and Bahia), and a destination’s frontier share is the fraction of its origin footprint located in frontier municipalities. For visualization, per-municipality embodied production can optionally be distributed onto 30 m MapBiomass soy pixels, each pixel of a municipality carrying that municipality’s per-pixel share for the given consumer country; this refinement is illustrative and does not affect the reported results.

²Stock additions and the balancing category are removed from final demand before aggregation: stock drawn down in year t embodies land use of earlier production years, and retaining it as negative year- t demand would attribute negative footprints to producer regions in drawdown years. EXIOBASE inventory changes are treated analogously.

1.9 Validation and benchmarking

TODO: FINISH

1.9.1 Benchmark and baseline

The reconstructed origin-to-importer flows X_{mj}^p (Section 1.6.1) are evaluated against the independent Trase Brazil–soy dataset (v2.6.1), which links municipalities of production to countries of first import using per-shipment customs and logistics records [7, 8]. As a second reference we construct a *pure-downscaling* baseline that uses no supply-chain structure at all: national exports to each destination are allocated to municipalities in proportion to their share of domestic-origin soybean supply (equal to production shares up to the treatment of stock changes), i.e. the allocation operator (1) applied directly to bilateral exports (Fig. 3b). The comparison thus places the model between a baseline without supply-chain information and a benchmark built from shipment-level records. All three sources are expressed in a common unit by converting oil and cake flows to bean equivalents with the national crush ratio (beans crushed per tonne of oil and cake produced). Trase flows with unknown municipality of origin are excluded, destinations are harmonized to the bilateral-trade classification, and the comparison sample comprises all soy-producing municipalities crossed with the destinations recording positive exports in both benchmark and model.

1.9.2 Comparison metrics

We compare modelled and benchmark flows on the municipality×destination matrix with three statistics: the Pearson correlation coefficient r (significance from the Fisher transform), the root-mean-square error, and the root-mean-square logarithmic error. Each is computed three ways: pooled over all flows, separately for each destination country, and after aggregating municipalities to states. A municipality-level assignment can also miss narrowly, placing a flow in a neighbouring municipality of the same production cluster. To separate such near-misses from genuine misallocation, we recompute all metrics on neighbourhood-smoothed flows under five spatial-weight schemes: first-order contiguity and a 100 km radius, each raw and row-standardized, plus row-standardized inverse distance. A conclusion that holds across all five schemes does not hinge on hitting the exact municipality.

1.9.3 Volume–correlation decision rule

Per-destination correlations are related to the destination’s total import volume to establish when subnational modelling adds information over pure downscaling. Small trade volumes are handled by few shipments and carry little spatial information, so per-destination agreement degrades for all methods below a volume floor (on the order of 10^3 t; Results). Figure 7 shows the flow-level pattern behind this rule for four sample years. The resulting decision rule (model subnationally above the floor, downscale below it) is reported with the benchmark tables produced for every study year.

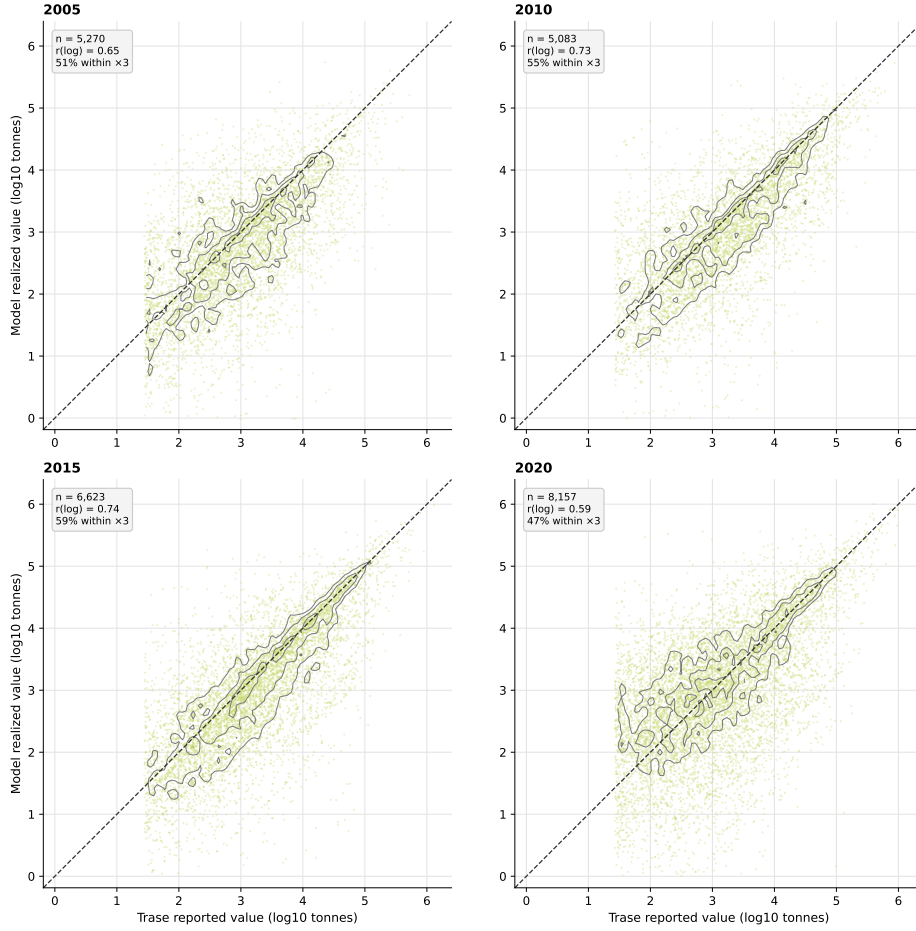


Fig. 7 Agreement with the benchmark grows with flow size (“rocket plots”). Every modelled municipality×destination soy export flow is plotted against the corresponding Trase flow (both $\log_{10}$ tonnes; flows below 1 t on either side are excluded) for four sample years, with kernel-density contours and the 1:1 line (dashed); each panel reports the number of matched flows, the log-scale Pearson correlation and the share of flows agreeing within a factor of three. Large flows converge onto the 1:1 diagonal (the “nose”), while small flows scatter (the “plume”): agreement rises systematically with volume, which motivates the volume floor of the decision rule — the tonnage that carries the footprint is also the best-validated part of the flow distribution.

1.9.4 Uncertainty and sensitivity

The pipeline is deterministic, so uncertainty is characterized by structured sensitivity rather than stochastic intervals, along four axes. (i) Spatial allocation tolerance: all agreement metrics are recomputed under the five neighbourhood-smoothing schemes of Section 1.9.2. (ii) Reference bracketing: every reported comparison places the model between the downscaling baseline and the shipment-based Trase benchmark, quantifying the value added by the supply-chain structure. (iii) Structural choices: the

two admissible treatments of net stock withdrawals (use-side proportional versus supply-side) are both implemented and were compared on a test year; the supply-side treatment is reported because it reflects the balance identity (a drawdown is a source of current-year supply) and alone keeps all municipal uses non-negative (Section 1.2.2). (iv) Backend vintage: the v1.1/v2 seam is quantified by running the 2010–2013 overlap years through both chains (Section 1.7.1); the -6 to -8% level difference and ≥ 0.988 cross-country correlation of the clean overlap years bound the vintage effect on pre-2010 results. Agreement with the benchmark rises strongly with flow size, so the largest flows, which carry most of the footprint, are also the best validated; results for individual small-volume destinations (below the $\sim 10^3$ t floor of Section 1.9.3) should be read with caution.

Finally, the proportional-mixing assumption of the trade-linkage inversion (Section 1.6.2) is probed by an origin-exclusion counterfactual for the EU, where segregated and certified sourcing is strongest. Setting the EU’s Amazon-biome soy sourcing to zero — a perfect Amazon Soy Moratorium — and redistributing that footprint across the EU’s remaining origins reallocates on average 0.9 Mha yr^{-1} (2010–2015) and moves the EU frontier-origin share from 28.8% to between 16% (displaced pro-rata to all non-Amazon origins) and 28.8% (displaced into the Cerrado frontier, which the moratorium does not cover); the all-destination footprint is unchanged, as this only redistributes within the EU consumption column. The bias is thus bounded and, notably, runs counter to the deforestation narrative: the model already attributes a higher Amazon-biome share to the EU (mean 14.8%) than to China (9.4%), so correcting toward observed EU segregation would lower, not raise, the EU’s frontier footprint.

1.10 Software

The pipeline is implemented in R (4.5; `data.table`, `dplyr`, `Matrix`, `sf`, `transport`) with figures and web-map tooling in Python (3.14; `pandas`, `matplotlib`, `geopandas`). The reported results have no proprietary software dependency; only the experimental multimodal-transport variant (not used here) requires a GAMS licence.

A Livestock system splits

Let s_m^k denote municipality m ’s share of system k in the 2010 gridded-livestock rasters (area-weighted zonal sums, normalized across the systems of each species); where a municipality reports a herd in the run year but had no gridded animals in 2010, the state-mean share is substituted. Pigs and chickens are split as

$$N_m^{\text{pig},k} = \text{pig}_m s_m^{\text{Pg},k}, \quad N_{m,\text{byd}}^{\text{chk}} = \text{chicken}_m s_m^{\text{Ch,Ext}}, \quad N_{m,\text{lay}}^{\text{chk}} = \text{chicken_layer}_m s_m^{\text{Ch,Int}}, \quad (10)$$

with broilers $N_{m,\text{bro}}^{\text{chk}} = (\text{chicken}_m - \text{chicken_layer}_m) s_m^{\text{Ch,Int}}$ (or $\text{chicken}_m s_m^{\text{Ch,Int}}$ where layers are unreported). Feedlot cattle are the 2006 census counts scaled by the national confinement growth factor for the run year (e.g. $4.38/3.46 \approx 1.27$ for 2013, from ABIEC/Athenagro), holding the municipal pattern fixed. The remaining cattle are split into grassland/mixed systems by the raster shares $s_m^{\text{Catt},u}$, $u \in \{\text{gra}, \text{mix}\}$, with

dairy identified by milked cows and meat as the residual:

$$N_{m,u,dair}^{cattle} = \text{cattle_milked}_m s_m^{Catt,u}, \quad (11)$$

$$N_{m,u,meat}^{cattle} = (\text{cattle}_m - \text{cattle_milked}_m - N_m^{cattle,flot}) s_m^{Catt,u}. \quad (12)$$

Buffalo use the municipal milked-cow ratio $\rho_m = \text{cattle_milked}_m / \text{cattle}_m$ as the dairy proxy,

$$N_{m,u,dair}^{buf} = \text{buffalo}_m \rho_m s_m^{Buff,u}, \quad N_{m,u,meat}^{buf} = \text{buffalo}_m (1 - \rho_m) s_m^{Buff,u}. \quad (13)$$

Each species' subsystems partition its reported herd exactly, e.g. $\text{cattle}_m = N_m^{cattle,flot} + \sum_u \sum_{v \in \{dair, meat\}} N_{m,u,v}^{cattle}$.

References

- [1] Bruckner, M. *et al.* FABIO—the construction of the food and agriculture biomass input–output model. *Environmental Science & Technology* **53**, 11302–11312 (2019).
- [2] Trsek, S. *Refining footprint analyses beyond the national level: An open tracing framework to capture the heterogeneity of subnational commodity flows*. Master's thesis, University of Natural Resources and Life Sciences (BOKU), Vienna (2022).
- [3] Schuhmacher, D. *et al.* transport: Computation of optimal transport plans and Wasserstein distances (2023). URL <https://cran.r-project.org/package=transport>. R package.
- [4] Kastner, T., Kastner, M. & Nonhebel, S. Tracing distant environmental impacts of agricultural products from a consumer perspective. *Ecological Economics* **70**, 1032–1040 (2011).
- [5] Miller, R. E. & Blair, P. D. *Input-Output Analysis: Foundations and Extensions* 2 edn (Cambridge University Press, Cambridge, 2009).
- [6] Stadler, K. *et al.* EXIOBASE 3: Developing a time series of detailed environmentally extended multi-regional input-output tables. *Journal of Industrial Ecology* **22**, 502–515 (2018).
- [7] Godar, J., Persson, U. M., Tizado, E. J. & Meyfroidt, P. Towards more accurate and policy relevant footprint analyses: Tracing fine-scale socio-environmental impacts of production to consumption. *Ecological Economics* **112**, 25–35 (2015).
- [8] Escobar, N. *et al.* Spatially-explicit footprints of agricultural commodities: Mapping carbon emissions embodied in Brazil's soy exports. *Global Environmental Change* **62**, 102067 (2020).